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SUPERSEDING
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MILITARY SPECIFICATION

REMOVER, AIRCRAFT CANOPY, M4 AND M5 METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Remover, Aircraft Canopy, M4 and M5, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550—Ammunition and Special Weapons, General Specification for
MIL-I-45208—Inspection System Requirements

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109—Quality Assurance Terms and Definitions
MIL-STD-413—Visual Inspection Guide for Rubber O-Rings
MIL-STD-639—Surface Roughness Preferred for Army Ordnance Materiel

DRAWINGS

MUNITIONS COMMAND

F 8595441—Remover, Aircraft Canopy, M4 and M5, Metal Parts Assembly
C 8823715—Container, Ammunition, Fiber M418 for Remover, Aircraft Canopy, M4 with Cartridge M29A2 or M5 with Cartridge M120
C 8823716—Box, Packing, Ammunition for Remover, Aircraft Canopy, M4 with Cartridge M29A2 or M5 with Cartridge M120
IEL 8595441—Index to Inspection Equipment List for Removers, Aircraft Canopy, M4 and M5, Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

FSC 1375

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3.1 General. Materials, parts and assemblies shall comply with all the requirements specified on the applicable drawings and specifications.

3.2 Initial production sample. Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Physical properties. Physical properties required by the drawing shall be uniform throughout each part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.5 Hydrostatic pressure. The components listed below shall withstand the corresponding internal pressure as indicated in table I for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.1).

TABLE I.

Component	Hydrostatic Pressure, (psi. minimum)
Outside Tube	3000
Telescoping Tube	1800
Inside Tube	2000

3.6 Shear pin strength. After application of protective finish, the pin shall shear when a load of 287 pounds plus or minus 28 pounds is applied at the steady rate of .25 inches per minute.

3.7 Structural strength. The metal parts assembly shall withstand a static load of 4000 pounds, minimum, for 15 seconds, minimum, in tension followed by a static load of 8000 pounds, minimum, for 15 seconds, minimum, in compression without permanent deformation or breakage of a part.

3.8 Torque. Torque requirements shall be as specified below. The run-on torque shall not be less than the indicated value noted in table II.

TABLE II.

Component	Run-on (inch pounds)
Cap and Outside Tube	75 min.
Sleeve and Outside Tube	75 min.
Firing Head and Inside Tube	75 min.

3.8.1 *Bottoming torque.* The bottoming torque between cap and outside tube and between sleeve and outside tube shall not exceed the running torque by 50 inch pounds. No bottoming torque shall be applied between firing head and inside tube.

3.9 Protective finish. Protective finish shall be in accordance with the applicable specification and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.10 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.11 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 *Metal defects.* All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.11.2 *Burr.* No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.11.3 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise

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specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to MIL–STD–109 to define the quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 **Initial production sample.** At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 5 metal parts assemblies, 10 sets of parts and fifty shear pins from the same lot of shear pins tested by the contractor. The samples shall be subjected to the tests in 4.3.3. Test procedures shall be in accordance with 4.4, except as provided in MIL–I–45208. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as used in regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of MIL–I–45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 **Inspection provisions.**

4.3.1 *Lot.*

4.3.1.1 *Submission of product.* The product shall be submitted in accordance with MIL–STD–105.

4.3.1.2 *Size of lot.* The shipping lots of metal parts assemblies shall consist of 616 removers, maximum, provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 *Examination.* One hundred per cent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in tables III and IV. For an individual defect (single characteristic) the criteria in table III shall be used. For total defects within a major or minor group the criteria in table IV shall be used. All non-conforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL–STD–413. The Ordnance Standard Finish Blocks prescribed in MIL–STD–639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE III.

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1– 300	100% Examination				
301– 1,300	110	1	2	2	3
1,301– 3,200	150	2	3	3	4
3,201– 8,000	225	3	4	4	5
8,000–22,000	300	4	5	5	6

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TABLE IV. *Total Allowable Defects per Sample Size*

Sample Sizes for Major Defects	110	150	225	300
No. of Defects in Class				
2– 3	1	2	3	4
4– 7	2	3	4	5
8–19	3	4	5	7
20–43	4	5	8	10
over 43	6	8	11	14
Sample Sizes for Minor Defects	110	150	225	300
No. of Defects in Class				
2– 3	2	3	4	5
4– 7	3	4	5	7
8–16	4	5	8	10
17–48	6	8	11	14
over 48	8	11	17	20

4.3.2.1 *Head, Firing, Dwg F8595442, covering a detail of Dwg F8595441.*

Defect	*Method of Inspection*
CRITICAL: None defined	
MAJOR:	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Pitch diameter of gas port thread, max.	Gage
104. Minor diameter of gas port thread, max.	Gage
105. Pitch diameter of set screw thread, max.	Gage
106. Minor diameter of set screw thread, max.	Gage
107. Concentricity of head with body, large end	Gage
108. Concentricity of midsection cavity with pitch diameter of thread	Gage
109. Concentricity of pitch diameter of thread with body, small end	Gage
110. Concentricity of body, small end with body, large end	Gage
111. Symmetry of elongated slot with midsection cavity	Gage
112. True position of pin hole with midsection cavity	Gage
113. True position of holes in flange with diameter of head and face of flange	Gage
114. True position of pitch diameter of gas port hole with midsection cavity	Gage
115. Diameter of undercut, max.	Gage
116. Diameter of midsection cavity	Gage
117. Diameter of pin hole, max.	Gage
118. Diameter of gas port counterbore	Gage
119. Diameter of body, large end	Gage
120. Diameter of body, small end	Gage
121. Diameter of chamfer at gas port	Gage
122. Depth to base of midsection cavity	Gage
123. Depth of threaded cavity including undercut	Gage
124. Depth of pin hole counterbore	Gage
125. Depth of gas port counterbore	Gage
126. Depth to end of thread, min.	Gage
127. Width of elongated slot, max.	Gage
128. Location of pin hole	Gage
129. Location of elongated slot	Gage
130. Location of gas port	Gage
131. Angular location of pin hole	Gage
132. Thread damaged or not full	Visual
133. Metal defective	Visual
134. Protective finish, inadequate or defective	Visual
135. Surface finish of gas port chamfer, counterbore and midsection cavity, improper	Visual
136. Burr or sharp corner	Visual
137. Foreign matter	Visual
MINOR:	
201. Pitch diameter of thread, min.	Gage
202. Pitch diameter of gas port thread, min.	Gage
203. Pitch diameter of set screw thread, min.	Gage
204. Diameter of head	Gage
205. Diameter of small cavity	Gage
206. Diameter of flange	Gage
207. Diameter of holes in flange	Gage
208. Diameter of pinhole, min.	Gage
209. Depth to bottom of drill point, max.	Gage
210. Depth to bottom of small cavity	Gage
211. Length to flange	Gage
212. Length to underside of flange	Gage
213. Length of body, small end	Gage
214. Length of elongated slot	Gage
215. Total length	Gage
216. Width across body at gas port flat	Gage
217. Width of elongated slot	Gage
218. Diameter of chamfer at pin hole	Measure

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Defect	Method of Inspection
219. Length of cutout on body, large end	Measure
220. Width of undercut, max.	Measure
221. Profile of elongated slot	Measure
222. Chamfer at mouth of midsection cavity	Measure
223. Marking missing, incorrect or illegible	Visual
224. Radius or chamfer, missing or incorrect	Visual
225. Surface finish, improper, except as otherwise classified	Visual

4.3.2.2 *Pin, Firing, Dwg C 8595443, covering a detail of Dwg F 8595441.*

Defect	Method of Inspection
CRITICAL:	
1. Diameter of striker, max.	Gage
MAJOR:	
101. Concentricity of groove with shoulder and flange	Gage
102. Concentricity of striker with shoulder and flange	Gage
103. True position of pin hole with shoulder and flange	Gage
104. True position of relief holes with shoulder and flange	Gage
105. Diameter of shoulder and flange	Gage
106. Diameter of pin hole, max.	Gage
107. Diameter of striker, min.	Gage
108. Diameter of groove	Gage
109. Depth of pin hole	Gage
110. Location of pin hole	Gage
111. Permissible variation of groove angles, max.	Gage
112. Metal defective	Visual
113. Protective finish inadequate or defective	Visual
114. Surface finish of shoulder, improper	Visual
115. Burr or sharp corner, except where permitted	Visual
116. Foreign matter	Visual
MINOR:	
201. Diameter of body	Gage
202. Diameter of relief holes	Gage
203. Diameter of pin hole, min.	Gage
204. Length from rear flange to face of shoulder	Gage
205. Total length	Gage
206. Length of shoulder	Gage
207. Width across flanges	Gage
208. Width of rear flange	Gage
209. Location of pin hole	Gage
210. Width of groove	Measure
211. Marking missing, incorrect or illegible	Visual
212. Radius or chamfer, missing or incorrect	Visual
213. Surface finish, improper, except as otherwise classified	Visual

4.3.2.3 *Pin, Shear, Dwg B8595444, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Diameter, max.	Gage
102. Total length	Gage
103. Metal defective	Visual
104. Protective finish, inadequate or defective	Visual
105. Surface finish, improper	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR:	
201. Profile of end	Measure

4.3.2.4 *Key, Locking, Dwg D8595445, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Symmetry of datum thickness with thickness	Gage
102. Thickness, min.	Gage
103. Thickness at datum, min.	Gage
104. Radius of small arc	Gage
105. Radius of large arc	Gage
106. Profile of chamfer ends	Gage
107. Metal defective	Visual
108. Protective finish, inadequate or defective	Visual
109. Surface finish of small arc, improper	Visual
110. Burr or sharp corner	Visual
111. Foreign matter	Visual
MINOR:	
201. Width	Gage
202. Thickness, Max.	Gage
203. Thickness at datum, max.	Gage
204. Marking missing, incorrect or illegible	Visual
205. Radius missing or incorrect	Visual
206. Surface finish, improper, except as otherwise classified	Visual

4.3.2.5 *Ring, Retaining, Dwg D8595446, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Concentricity of groove with inside diameter	Gage
102. Concentricity of outside diameter with inside diameter	Gage
103. Symmetry of groove width at datum with width	Gage
104. Diameter, outside, Min.	Gage
105. Diameter, inside, max.	Gage
106. Diameter of groove	Gage
107. Width of groove at datum	Gage

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Defect	Method of Inspection
108. Profile of groove	Gage
109. Metal defective	Visual
110. Protective finish, inadequate or defective	Visual
111. Surface finish of inside diameter, improper	Visual
112. Burr or sharp corner	Visual
113. Foreign matter	Visual
MINOR:	
201. Diameter, outside, max.	Gage
202. Diameter, inside, min.	Gage
203. Width	Gage
204. Height	Gage
205. Marking missing, incorrect or illegible	Visual
206. Radius or chamfer, missing or incorrect	Visual
207. Surface finish, improper, except as otherwise classified	Visual

4.3.2.6 *Sleeve, Dwg C8595447, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of large bore with pitch diameter of thread	Gage
104. Concentricity of small bore with large bore	Gage
105. Symmetry of width of flats with body diameter	Gage
106. Diameter of large bore, max.	Gage
107. Diameter of small bore	Gage
108. Diameter of undercut, max.	Gage
109. Thread damaged or not full	Visual
110. Metal defective	Visual
111. Protective finish inadequate or defective	Visual
112. Burr or sharp corner	Visual
113. Foreign matter	Visual
MINOR:	
201. Pitch diameter of thread, min.	Gage
202. Diameter of large bore, min.	Gage
203. Diameter of body	Gage
204. Depth to small bore	Gage
205. Depth to large bore	Gage
206. Width across flats	Gage
207. Width of undercut, max.	Measure
208. Total length	Measure
209. Length of flats	Measure
210. Marking missing, incorrect or illegible	Visual
211. Radius or chamfer, missing or incorrect	Visual
212. Surface finish, improper	Visual

4.3.2.7 *Washer, Dwg C8595448, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Concentricity of outside diameter with inside diameter	Gage
102. Diameter, outside, min.	Gage
103. Diameter, inside, max.	Gage
104. Metal defective	Visual
105. Protective finish, inadequate or defective	Visual
106. Burr or sharp corner	Visual
107. Foreign matter	Visual
MINOR:	
201. Diameter, outside, max.	Gage
202. Diameter, inside, min.	Gage
203. Thickness	Gage
204. Marking missing, incorrect or illegible	Visual
205. Surface finish, improper	Visual

4.3.2.8 *Ring, Sealing, Dwg C8595449, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Diameter, inside, max.	Gage
102. Thickness	Gage
103. Width	Gage
104. Material defective	Visual
105. Flash	Visual
106. Foreign matter	Visual
MINOR:	
201. Diameter, inside, min.	Gage

4.3.2.9 *Stop, Firing Pin, Dwg D8595450, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL:	
1. Diameter of cavity, min.	Gage
2. Diameter of striker hole, min.	Gage
MAJOR:	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of striker hole with pitch diameter of thread	Gage
104. Concentricity of cavity with pitch diameter of thread	Gage
105. Concentricity of datum diameter with pitch diameter of thread	Gage
106. True position of holes with pitch diameter of thread	Gage
107. Diameter of cavity, max.	Gage
108. Depth of cavity, min.	Gage
109. Location of datum diameter	Gage
110. Height of boss	Gage
111. Angle of chamfer on boss	Gage
112. Thread damaged or not full	Visual

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Defect	Method of Inspection
113. Metal defective	Visual
114. Protective finish, inadequate or defective	Visual
115. Burr	Visual
116. Foreign matter	Visual
MINOR:	
201. Pitch diameter of thread, max.	Gage
202. Diameter of striker hole, max.	Gage
203. Diameter of hole	Gage
204. Depth of cavity, max.	Gage
205. Total length	Gage
206. Marking missing, incorrect or illegible	Visual
207. Radius or chamfer, missing or incorrect	Visual
208. Surface finish, improper	Visual

4.3.2.10 *Tube, Inside, Dwg F8595452, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Pitch diameter o fthread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of pitch diameter of thread with inside diameter to datum location	Gage
104. Concentricity of groove with shoulder to datum location	Gage
105. Diameter of shoulder, groove end	Gage
106. Diameter of shoulder, thread end	Gage
107. Diameter, inside, max.	Gage
108. Diameter of groove	Gage
109. Length of effective thread, min.	Gage
110. Total length, max.	Gage
111. Thickness of body wall, min.	Gage
112. Thickness of shoulder wall, min.	Gage
113. Straightness	Gage
114. Radius of inside diameter, threaded end	Gage
115. Thread damaged or not full	Visual
116. Metal defective	Visual
117. Protective finish, inadequate or defective	Visual
118. Burr or sharp corner	Visual
119. Foreign matter	Visual
MINOR:	
201. Pitch diameter of thread, max.	Gage
202. Diameter, inside, min.	Gage
203. Diameter of body	Gage
204. Total length, min.	Gage
205. Width of groove	Gage
206. Location of groove	Gage
207. Length of shoulder, groove end	Measure
208. Length of shoulder, thread end	Measure
209. Length of thread, max.	Measure
210. Marking missing, incorrect or illegible	Visual
211. Radius or chamfer, missing or incorrect	Visual
212. Surface finish, improper	Visual

4.3.2.11 *Ring, Sealing, Dwg C8595453, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Diameter, inside, max.	Gage
102. Thickness, min.	Gage
103. Material defective	Visual
104. Flash	Visual
105. Foreign matter	Visual
MINOR:	
201. Diameter, inside, min.	Gage
202. Width	Gage
203. Thickness, max.	Gage

4.3.2.12 *Tube, Telescoping, Dwg F8595454, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL:	
1. Diameter of body, max.	Gage
MAJOR:	
101. Concentricity of groove with shoulder	Gage
102. Concentricity of shoulder with inside diameter to datum	Gage
103. Diameter of body, min.	Gage
104. Diameter of shoulder, min.	Gage
105. Diameter, inside, max.	Gage
106. Diameter of groove	Gage
107. Total length, max.	Gage
108. Straightness	Gage
109. Variation in wall thickness	Gage
110. Metal defective	Visual
111. Protective finish, inadequate or defective	Visual
112. Surface finish of body and inside diameter, improper	Visual
113. Burr or sharp corner	Visual
114. Foreign matter	Visual
MINOR:	
201. Diameter, inside, min.	Gage
202. Diameter of shoulder, max.	Gage
203. Length of shoulder	Gage
204. Total length, min.	Gage
205. Width of groove	Gage
206. Location of groove	Gage
207. Chamfer, incorrect	Measure
208. Marking missing, incorrect or illegible	Visual
209. Radius missing or incorrect	Visual
210. Surface finish, improper, except as otherwise classified	Visual

4.3.2.13 *Ring, Sealing, Dwg C8595455, covering a detail of Dwg F8595441.*

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Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Diameter, inside, max.	Gage
102. Thickness, min.	Gage
103. Material defective	Visual
104. Flash	Visual
105. Foreign matter	Visual
MINOR:	
201. Diameter, inside, min.	Gage
202. Width	Gage
203. Thickness, Max.	Gage

4.3.2.14 *Tube, Outside, Dwg F8595456, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Pitch diameter of thread, min. (both ends)	Gage
102. Major diameter of thread, min. (both ends)	Gage
103. Concentricity of pitch diameter of thread (both ends) with inside diameter to datum location	Gage
104. Concentricity of counterbore with pitch diameter of thread	Gage
105. Concentricity of chamfer at datum with pitch diameter of thread	Gage
106. Diameter of counterbore	Gage
107. Diameter, inside, max.	Gage
108. Depth of counterbore	Gage
109. Total length, min.	Gage
110. Length of effective thread, min.	Gage
111. Length of chamfer	Gage
112. Thickness of wall, min.	Gage
113. Location of datum diameter at chamfer	Gage
114. Straightness	Gage
115. Angle of chamfer	Gage
116. Thread damaged or not full	Visual
117. Metal defective	Visual
118. Protective finish, inadequate or defective	Visual
119. Surface finish of chamfer and inside diameter, improper	Visual
120. Burr or sharp corner	Visual
121. Foreign matter	Visual
MINOR:	
201. Pitch diameter of thread, max.	Gage
202. Diameter, inside, min.	Gage
203. Diameter of body	Gage
204. Total length, max.	Gage
205. Length of chamfer, min.	Gage
206. Length to end of shoulder (both ends)	Measure
207. Length of thread, max. (both ends)	Measure
208. Marking missing, incorrect or illegible	Visual
209. Radius or chamfer, missing or incorrect except as otherwise classified	Visual
210. Surface finish, improper, except as otherwise classified	Visual

4.3.2.15 *Gasket, Dwg C8595457, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Diameter, inside	Gage
102. Width, min.	Gage
103. Thickness, min.	Gage
104. Material defective	Visual
105. Flash	Visual
106. Foreign matter	Visual
MINOR:	
201. Width, max.	Gage
202. Thickness, Max.	Gage

4.3.2.16 *Cap, Dwg F8595458, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of body with pitch diameter of thread	Gage
104. Symmetry of lug with pitch diameter of thread	Gage
105. True position of mounting hole with thread including perpendicularity with lug	Gage
106. Diameter of undercut, max.	Gage
107. Thread damaged or not full	Visual
108. Metal defective	Visual
109. Protective finish, inadequate or defective	Visual
110. Burr or sharp corner	Visual
111. Foreign matter	Visual
MINOR:	
201. Pitch diameter of thread, min.	Gage
202. Diameter of mounting hole	Gage
203. Diameter of body	Gage
204. Diameter of undercut, min.	Gage
205. Depth of cavity	Gage
206. Length of body	Gage
207. Total length	Gage
208. Width of undercut	Gage
209. Thickness of lug	Gage
210. Location of mounting hole	Gage
211. Length to taper of lug	Measure
212. Profile of lug	Measure
213. Marking missing, incorrect or illegible	Visual
214. Radius or chamfer, missing or incorrect	Visual
215. Surface finish, improper	Visual

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4.3.2.17 *Tube, Outside and Pellet Assembly, Dwg D8595460, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Protrusion of pellet above thread	Gage
102. Angular location of pellet	Gage
103. Pellet missing, loose or improperly inserted	Manual/ Visual
104. Thread damaged	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR:	
201. Location of pellet	Gage
202. Pellet radius, incorrect	Visual

4.3.2.18 *Tube, Inside and Pellet Assembly, Dwg D10521228, covering a detail of Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Protrusion of pellet above thread	Gage
102. Angular location of pellet	Gage
103. Pellet missing, loose or improperly inserted	Manual/ Visual
104. Thread damaged	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR:	
201. Location of pellet	Gage
202. Pellet radius, incorrect	Visual

4.3.2.19 *Metal Parts Assembly, Dwg F8595441.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR:	
101. Any part missing, except as otherwise classified	Visual
102. Improper assembly	Visual
103. Thread damaged	Visual
104. Marking missing, incorrect or illegible	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR:	
201. Total length, max.	Gage
202. Shipping plug missing	Visual
203. Protective finish, inadequate or defective	Visual

4.3.2.20 *Fiber Container, Dwg C8823715.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR: None defined	
MINOR:	
201. Contents loose	Manual
202. Metal ends loose or distorted	Manual
203. Gap between body and cover, max.	Visual
204. Cut or damage through all impregnated layers	Visual

4.3.2.21 *Packing Box, Dwg C8823716.*

Defect	Method of Inspection
CRITICAL: None defined	
MAJOR: None defined	
MINOR:	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual/ Visual
203. Strapping missing, loose, broken or improperly assembled	Manual/ Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

4.3.3 *Inspection testing.* The following tests shall be performed in accordance with the provisions of 4.4 except as provided in MIL–I–45208.

4.3.3.1 *Hardness.* Fifty units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.3.3.2 *Hydrostatic pressure.* Each inside, telescoping and outside tube shall be subjected to the internal hydrostatic pressure test. Failure of any tube to comply with the requirements of 3.5 shall be cause for rejection of the tube.

4.3.3.3 *Shear pin strength.* Fifty pins from each lot shall be tested for shear strength. Failure of any pin to comply with the requirement of 3.6 shall be cause for rejection of the lot.

4.3.3.4 *Structural strength.* Fifteen removers from each lot shall be tested for structural strength. Failure of any remover to comply with the requirement of 3.7 shall be cause for rejection of the lot.

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4.3.3.5 *Protective finish.* The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standard listed on the drawing.

4.3.3.6 *Residual magnetism.* Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirements of 3.10 shall be cause for rejection of the assembly.

4.3.4 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL–8595441, except as provided in MIL–I–45208.

4.4 **Test methods and procedures.**

4.4.1 *Hardness.* Prior to the application of protective finish, a hardness tester shall be used to determine the hardness values specified on the detail drawing.

4.4.2 *Hydrostatic pressure.* Prior to the application of protective finish, each inside, telescoping and outside tube shall be mounted in the device and subjected to an internal hydrostatic pressure to determine compliance with 3.5.

4.4.3 *Shear pin strength.* After application of protective finish, the pin shall be subjected to the shear pin strength test. The test shall be performed in the fixture in conjunction with a tensile testing machine.

4.4.4 *Structural strength.* The remover shall be placed in a tensile testing machine and subjected to the specified structural loads.

4.4.5 *Protective finish.* The test procedure for protective finish shall be in accordance with the applicable Military Standard referenced on the drawing.

4.4.6 *Residual magnetism.* The remover shall be tested in an area free of local magnetic effects by placing the remover 6 inches from and at the same height as the compass, in a north-south horizontal position (firing head end at south) with the compass set in an east-west heading as shown in figure 1. The remover shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass.

5. PREPARATION FOR DELIVERY

5.1 **Packing and marking.** Level A—Packing and marking shall be in accordance with Drawings C8823715 and C8823716.

5.2 **Data cards.** Data card information shall be as specified in MIL–A–2550.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

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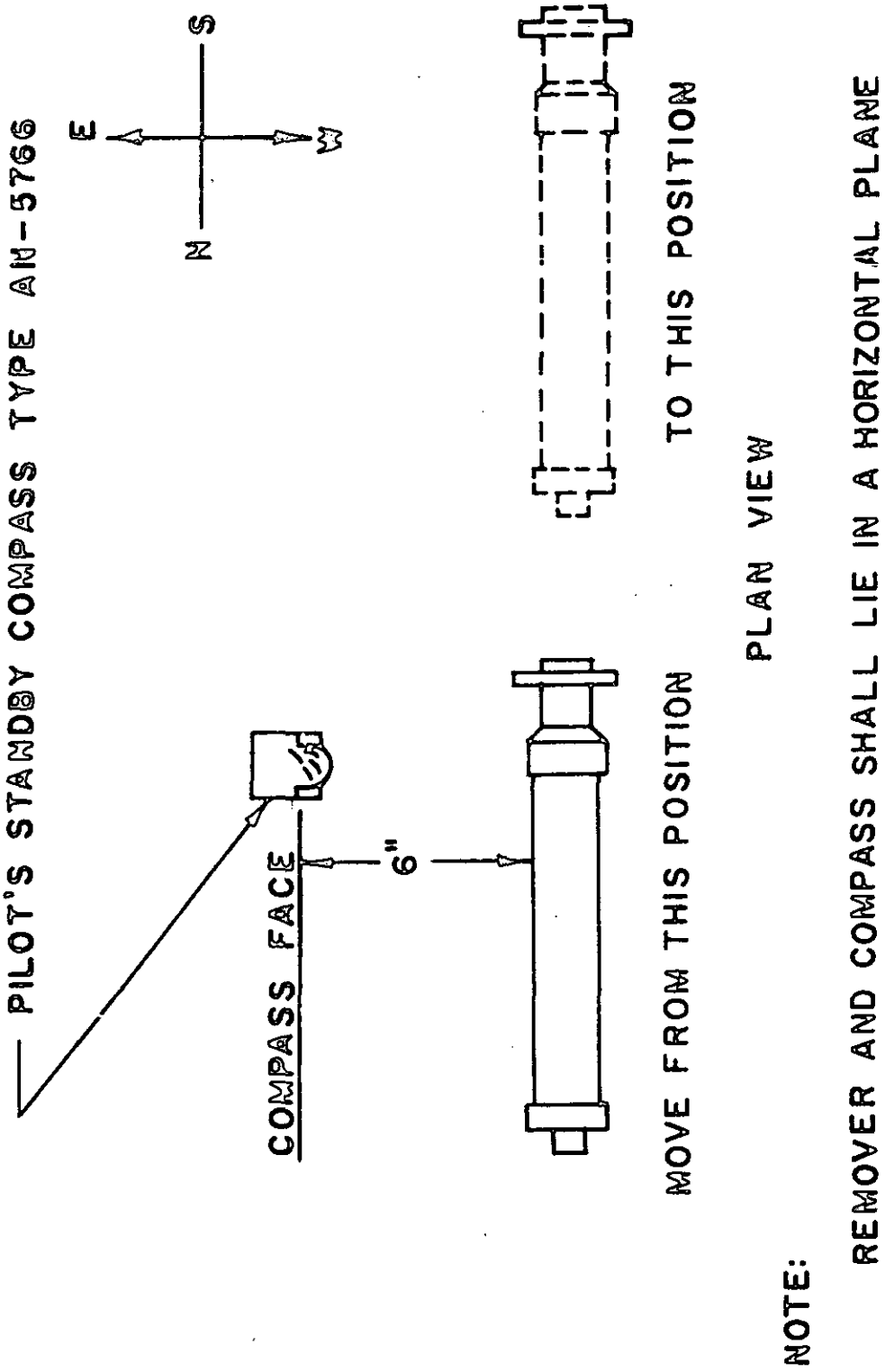


FIGURE 1. *Layout for determination of residual magnetism in remover, aircraft canopy, M4 and M5.*

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